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Exp No: 10	Internetworking with routers in CISCO PACKET TRACER simulator

AIM:

To design and configure internetworks in Cisco Packet Tracer using (a) a router with static IP addresses, (b) a wireless router with DHCP server and an internet cloud, and (c) a switch, a router and Ethernet cables in the lab, and to trace how the packets are routed.

ALGORITHM / PROCEDURE:

(a) Simple internetwork using a router

1. Place a Router (2911), two Switches and six PCs (3 in each network) on the workspace.
2. Connect each PC to its switch and each switch to the router with **copper straight-through cables**. Router interface `Gig0/0` goes to Switch0 and `Gig0/1` goes to Switch1.
3. Assign the static IP addresses to the PCs (network 1: `192.168.1.0/24`, network 2: `192.168.2.0/24`) and set the **default gateway** of each PC to the router interface address (`192.168.1.1` / `192.168.2.1`).
4. Open the router CLI and configure both interfaces with `ip address` and `no shutdown` (commands given below).
5. Ping from PC0 to PC3 and check the reply. Use `tracert` to see the path.
6. Go to Simulation mode, add a simple PDU from PC0 to PC3 and trace the packet through Switch0, Router0 and Switch1.

(b) Wireless router, DHCP server and internet cloud

1. Place a Wireless Router, a Cable Modem, a Cloud, a Server, one PC (wired), a Laptop and a Smartphone.
2. Connect PC0 to a LAN port of the wireless router, the Internet port of the router to the Cable Modem, the modem to the Cloud and the Cloud to the Server (`Coaxial / Ethernet`).
3. Open the wireless router **GUI**: set **SSID = LabNet**, **WPA2-PSK password** and **Internet setup = DHCP**. Set the **DHCP server = Enabled** (start IP `192.168.0.100`, 50 users).
4. Give the server a static IP `203.0.113.10`. On the Laptop and the Smartphone configure the **Wireless0** as SSID `LabNet` with the same password.
5. In IP Configuration select **DHCP** on the PC, Laptop and Smartphone and check the address received.
6. Ping the server and also open its web page from the browser of the Laptop.

(c) Inter-network in the lab using switch, router and Ethernet cables

1. Take one router, two switches, six PCs and eight straight-through UTP cables (made in Experiment 2).
2. Connect PCs to the switches, and the switches to the two Gigabit ports of the router. Check that all the link LEDs glow.
3. Assign the addresses `172.16.1.0/24` and `172.16.2.0/24` with the router as the default gateway and configure the router interfaces.
4. Ping within the same network and between the two networks. Note down the results.

Connection	Cable used
PC to Switch	Straight-through
Switch to Router	Straight-through
PC to Router (direct)	Crossover

Connection	Cable used
Router to Router (LAN side)	Crossover

Router configuration (Part a)

Router CLI

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#copy running-config startup-config
```

DIAGRAM:

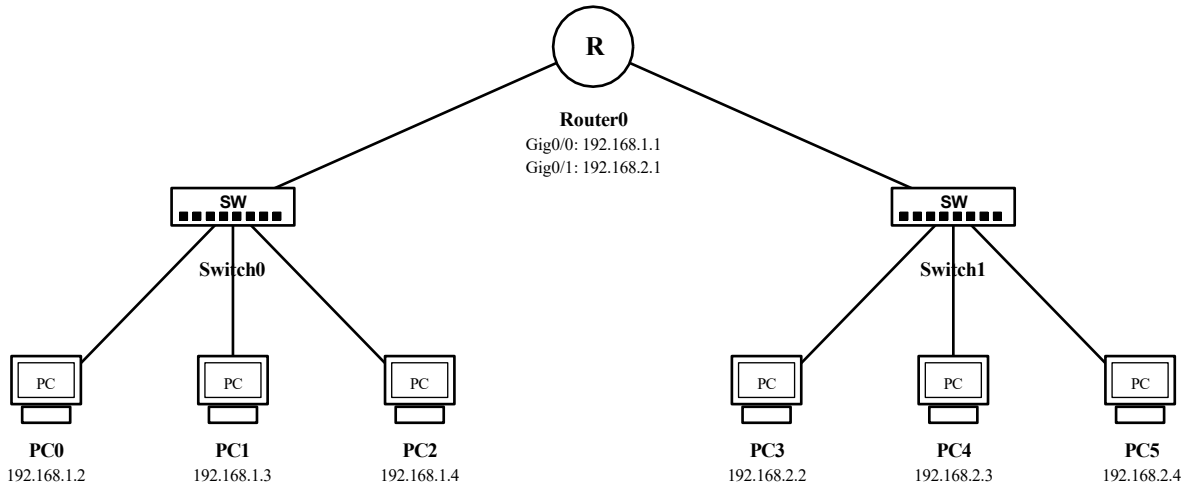


Fig 1: Internetwork of two LANs connected by a router (Part a)

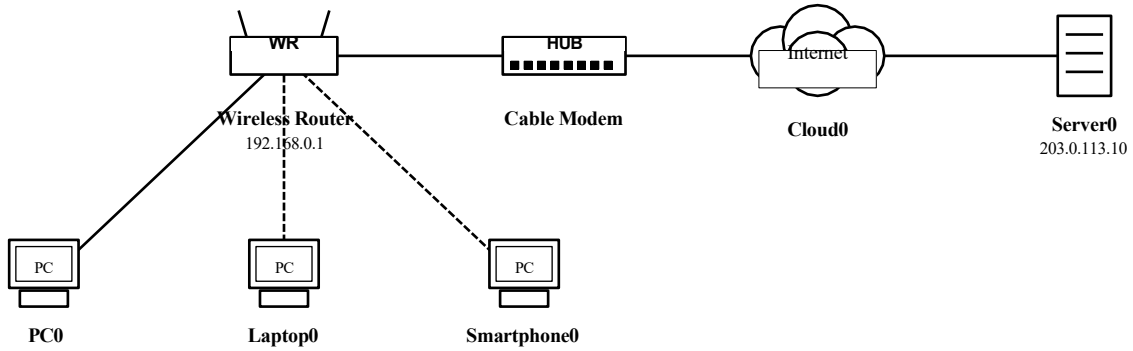


Fig 2: Wireless router with DHCP and internet cloud (Part b) - dotted lines are wireless links

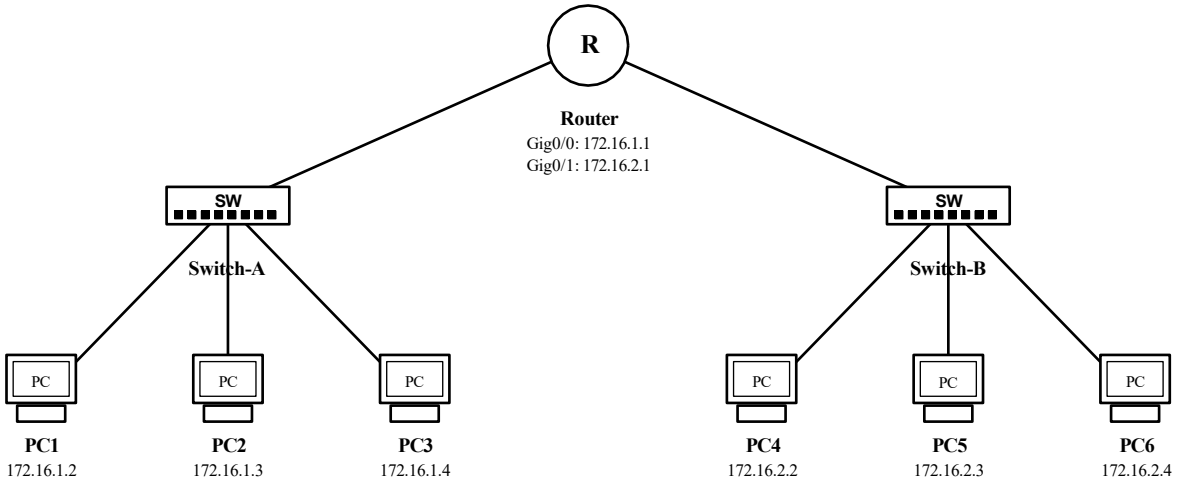
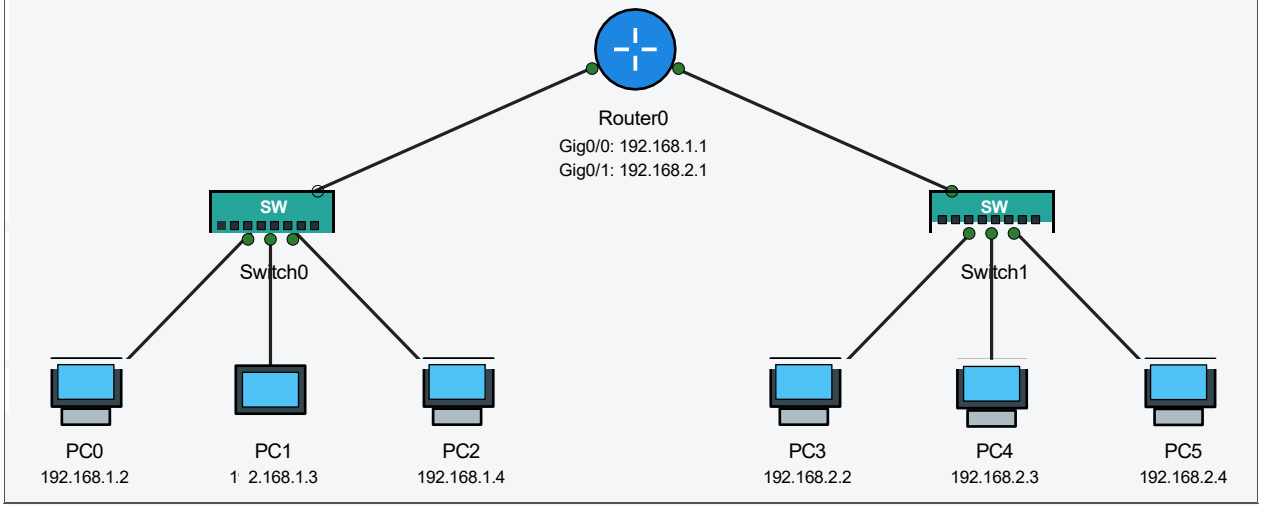


Fig 3: Lab inter-network using switches, router and Ethernet cables (Part c)

OUTPUT:

Part (a) - Router with two networks

Cisco Packet Tracer - internetwork_router.pkt



Packet Tracer view of Part (a)

R1 - CLI

```
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(config-if)#end
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES  manual  up                  up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
```

R1 - CLI (contd.)

```
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
```

PC0 - Command Prompt

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=2ms TTL=127
Reply from 192.168.2.3: bytes=32 time=1ms TTL=127
Reply from 192.168.2.3: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>tracert 192.168.2.3

Tracing route to 192.168.2.3 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    0 ms      0 ms      1 ms      192.168.2.3

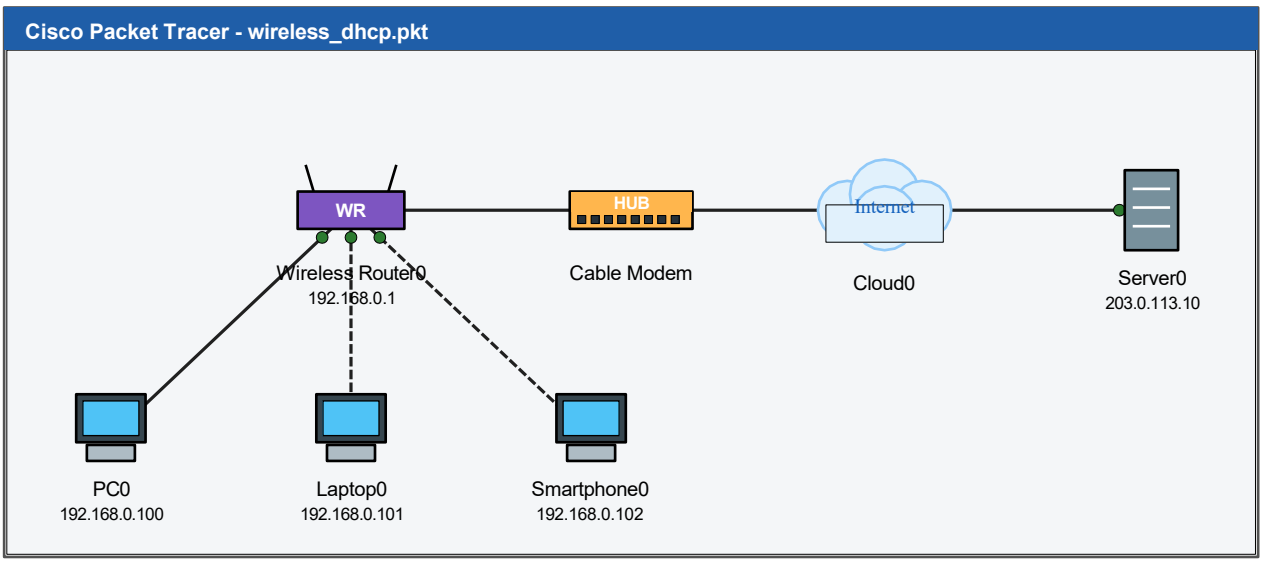

Trace complete.
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Switch0	ICMP		0.000	N	0
	Successful	Switch0	Router0	ICMP		0.001	N	1
	Successful	Router0	Switch1	ICMP		0.002	N	2
	Successful	Switch1	PC3	ICMP		0.003	N	3
	Successful	PC3	Switch1	ICMP		0.004	N	4
	Successful	Switch1	Router0	ICMP		0.005	N	5
	Successful	Router0	Switch0	ICMP		0.006	N	6
	Successful	Switch0	PC0	ICMP		0.007	N	7

The router receives the frame on Gig0/0, removes the Ethernet header, looks at the destination IP in the routing table, and sends a new frame out of Gig0/1 to PC3.

Part (b) - Wireless router, DHCP and internet cloud

Cisco Packet Tracer - wireless_dhcp.pkt



Packet Tracer view of Part (b)

Laptop0 - Command Prompt

```
C:\>ipconfig /all

Wireless0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address.................: 0001.C7AB.4D12
   Link-local IPv6 Address.........: FE80::201:C7FF:FEAB:4D12
   IPv6 Address....................: ::
   IP Address......................: 192.168.0.101
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.0.1
   DHCP Servers....................: 192.168.0.1
   DNS Servers.....................: 0.0.0.0

C:\>ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Reply from 203.0.113.10: bytes=32 time=21ms TTL=126
Reply from 203.0.113.10: bytes=32 time=12ms TTL=126
Reply from 203.0.113.10: bytes=32 time=14ms TTL=126
Reply from 203.0.113.10: bytes=32 time=11ms TTL=126

Ping statistics for 203.0.113.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 21ms, Average = 14ms
```

Device	Connection	Method	IP address received	Ping to server
PC0	Wired (LAN port)	DHCP	192.168.0.100	Successful
Laptop0	Wireless (LabNet)	DHCP	192.168.0.101	Successful
Smartphone0	Wireless (LabNet)	DHCP	192.168.0.102	Successful

Part (c) - Lab network with switches, router and Ethernet cables

PC1 - Command Prompt

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   IP Address......................: 172.16.1.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 172.16.1.1

C:\>ping 172.16.1.3

Pinging 172.16.1.3 with 32 bytes of data:

Reply from 172.16.1.3: bytes=32 time=1ms TTL=127
Reply from 172.16.1.3: bytes=32 time=<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time=<1ms TTL=127
Reply from 172.16.1.3: bytes=32 time=<1ms TTL=127

Ping statistics for 172.16.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 172.16.2.4

Pinging 172.16.2.4 with 32 bytes of data:

Request timed out.
Reply from 172.16.2.4: bytes=32 time=2ms TTL=127
Reply from 172.16.2.4: bytes=32 time=1ms TTL=127
Reply from 172.16.2.4: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.2.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

Source	Destination	Same network?	Through router?	Result
PC1 (172.16.1.2)	PC3 (172.16.1.4)	Yes	No	Successful
PC1 (172.16.1.2)	PC4 (172.16.2.2)	No	Yes	Successful
PC6 (172.16.2.4)	PC2 (172.16.1.3)	No	Yes	Successful

RESULT:

Thus the internetworks were designed and configured in Cisco Packet Tracer using a router with static IP addresses, a wireless router with DHCP server and internet cloud, and a lab network using switches, a router and Ethernet cables. The packet transmission across the networks was traced in simulation mode.